



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI PERUMALAPPAN FUELS UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

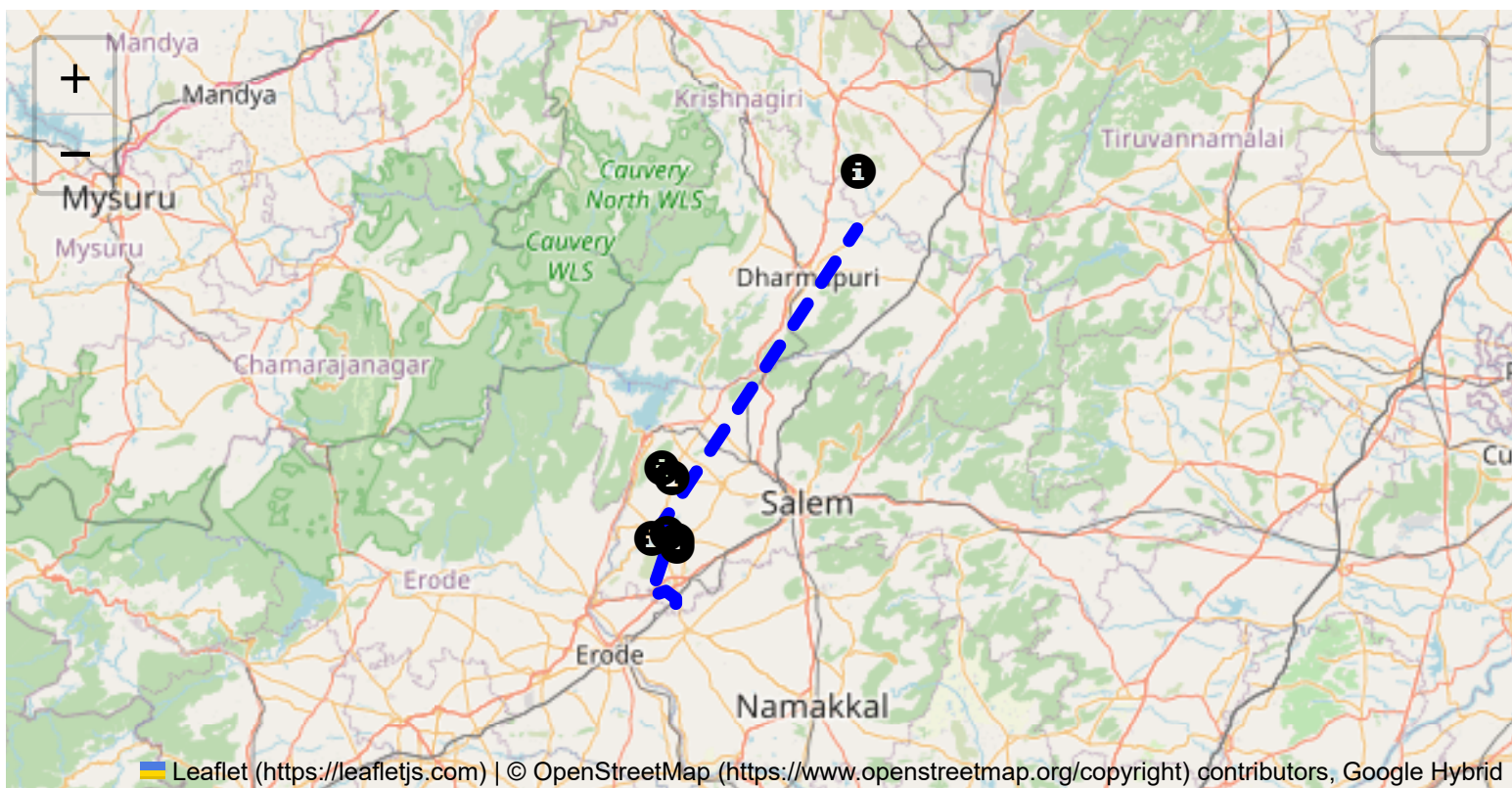
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

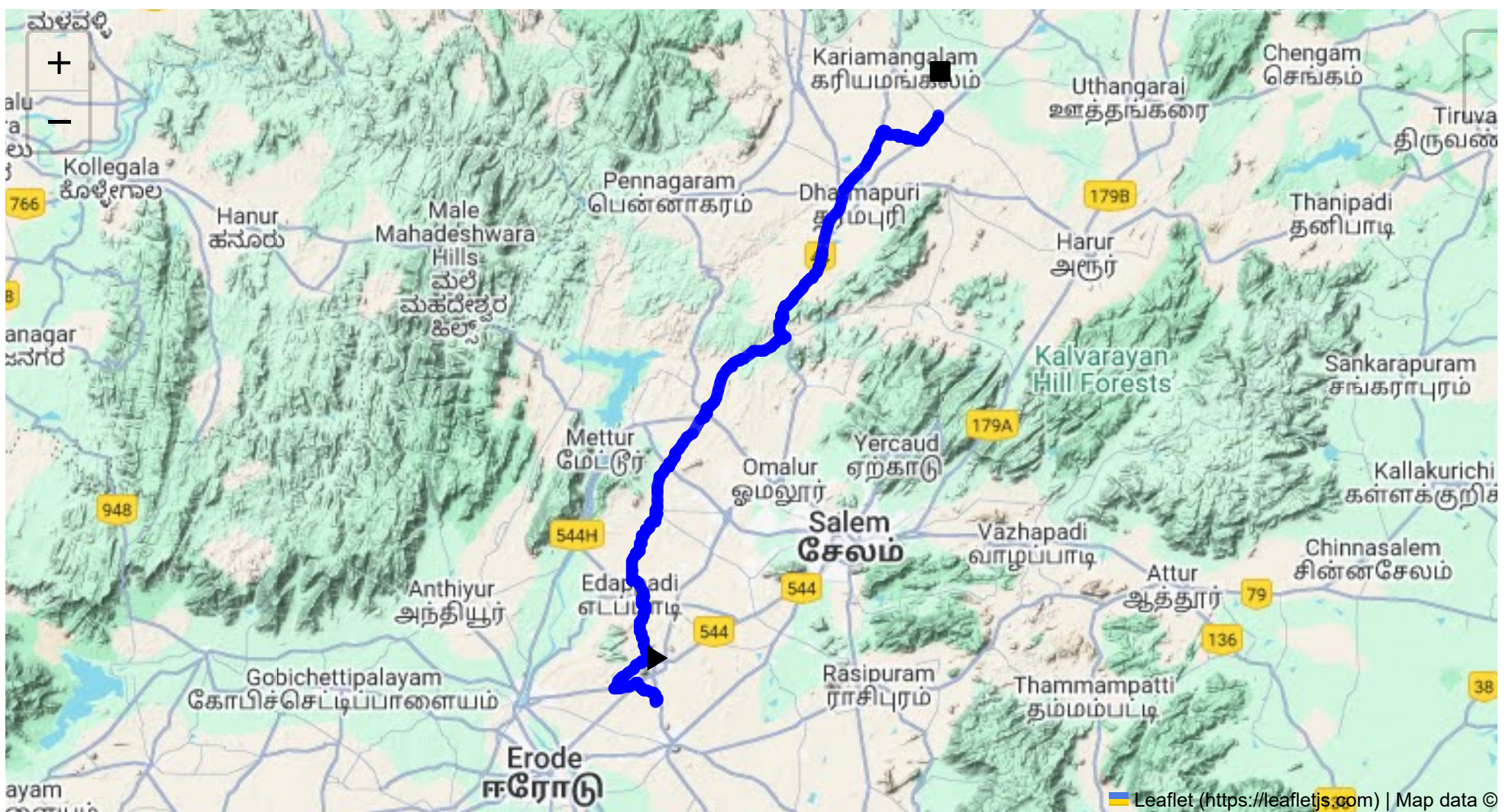
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 126.75 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.2400137, 78.2685629)



Welcome to the Journey Risk Management Study

Route Overview: The route from CVQF+23W, Sangagiri to 67Q9+VFR, Bannikulam via the intermediate points stretches approximately 127 kilometers through Tamil Nadu. It primarily follows regional and state highways, with varying road types from well-paved main roads to potentially narrower local roads.

Typical Weather Conditions: Tamil Nadu generally experiences a tropical climate. The area might see hot and humid conditions year-round, with temperatures rising significantly during summer months.

Monsoon season (June to September) can introduce heavy rains, causing potential flooding and slippery road conditions which may impair visibility and traction.

Traffic Patterns: Traffic congestion is prevalent near major towns and intersections, particularly during morning (8-10 AM) and evening (5-7 PM) rush hours. The areas around Sankari and Bhavani Main Road experience regular congestion, given their connectivity to larger commercial hubs.

Road Quality and Infrastructure: The main roads are mostly in good condition, though some segments might be affected by potholes and uneven surfaces, particularly during and after the monsoon season. Local roads could be narrower, with less maintenance, challenging navigation for larger vehicles.

Alternate Routes for Emergencies: Potential alternate routes include bypass roads through less congested rural regions, though detailed mapping and verification for truck suitability are essential. Local advice should be sought for optimal emergency detours, considering the road quality.

Local Regulations on Hazardous Material Transport: Transport of hazardous materials is strictly regulated with requirements for permits and adherence to designated routes. Ensure compliance with all state transport regulations and maintain documentation for inspections.

Historical Incidents: There’s a limited record of incidents specific to hazardous materials in this area. However, the risk of accidents tends to increase with poor weather and congested conditions.

Environmental Considerations: Potential environmental concerns include agricultural areas which may require extra precautions to prevent contamination. These areas are economically significant and environmentally sensitive, necessitating careful driving to avoid spills or leaks.

Communication Coverage: Mobile network coverage is generally sufficient along main roads, though rural sections may experience dead zones. It is advisable to equip vehicles with GPS devices with offline maps and satellite communication capabilities if necessary.

Emergency Response Times: Urban areas have quicker response times, often within 30-45 minutes. Rural segments could experience delays of up to 1-2 hours due to distance from major emergency services.

Overall Summary: This route poses moderate risk due to traffic congestion, weather variability, and infrastructure quality. Ensuring proper planning—allowing flexibility due to weather, traffic, and road conditions—is essential for safe transport. Awareness of local regulations, alternative routes, and emergency strategies will enhance safety. Frequent communication with local authorities and maintenance of safety protocols is strongly advised to mitigate risks associated with hazardous material transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60359, 77.83965	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
2	Roundabout	High	12.23740, 78.27029	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.44900, 77.87406	15 KM/Hr
6	Turn	High	11.45352, 77.85698	15 KM/Hr
7	Turn	Medium	11.45764, 77.85219	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	High	11.45571, 77.81702	15 KM/Hr
10	Turn	High	11.45615, 77.81677	15 KM/Hr
11	Turn	Medium	11.46916, 77.82431	30 KM/Hr
12	Turn	Medium	11.49385, 77.86709	30 KM/Hr
13	Turn	High	11.49404, 77.86724	15 KM/Hr
14	Turn	Medium	11.49513, 77.86656	30 KM/Hr
15	Turn	Medium	11.50961, 77.86008	30 KM/Hr
16	Turn	Medium	11.52587, 77.85524	30 KM/Hr
17	Turn	Medium	11.52791, 77.85581	30 KM/Hr
18	Turn	Medium	11.55685, 77.85813	30 KM/Hr
19	Turn	Medium	11.56411, 77.85969	30 KM/Hr
20	Turn	High	11.57903, 77.85441	15 KM/Hr
21	Turn	Medium	11.57910, 77.85555	30 KM/Hr
22	Turn	High	11.58022, 77.85744	15 KM/Hr
23	Turn	Medium	11.58709, 77.85824	30 KM/Hr
24	Blind Spot	Blind Spot	11.60437, 77.83983	10 KM/Hr
25	Turn	Medium	11.61007, 77.84163	30 KM/Hr
26	Turn	Medium	11.61835, 77.84026	30 KM/Hr
27	Turn	Medium	11.63147, 77.84083	30 KM/Hr
28	Turn	Medium	11.64049, 77.84596	30 KM/Hr
29	Turn	Medium	11.64166, 77.84871	30 KM/Hr
30	Turn	Medium	11.64516, 77.85108	30 KM/Hr
31	Turn	Medium	11.65359, 77.85231	30 KM/Hr
32	Turn	High	11.67319, 77.86041	15 KM/Hr
33	Turn	Medium	11.67365, 77.86215	30 KM/Hr
34	Turn	Medium	11.68487, 77.86933	30 KM/Hr
35	Turn	High	11.68478, 77.87285	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	High	11.69463, 77.87589	15 KM/Hr
37	Turn	High	11.69549, 77.87444	15 KM/Hr
38	Turn	High	11.76020, 77.89071	15 KM/Hr
39	Turn	High	11.75960, 77.89221	15 KM/Hr
40	Turn	Medium	11.83330, 77.94432	30 KM/Hr
41	Turn	Medium	11.83370, 77.94485	30 KM/Hr
42	Turn	Medium	11.84179, 77.94732	30 KM/Hr
43	Turn	Medium	11.89747, 77.98079	30 KM/Hr
44	Turn	Medium	11.90011, 77.98213	30 KM/Hr
45	Turn	Medium	11.91901, 78.02623	30 KM/Hr
46	Turn	High	11.93618, 78.05392	15 KM/Hr
47	Turn	Medium	11.93710, 78.05390	30 KM/Hr
48	Turn	High	11.95824, 78.04981	15 KM/Hr
49	Turn	Medium	11.95992, 78.04947	30 KM/Hr
50	Turn	Medium	11.96320, 78.05139	30 KM/Hr
51	Turn	High	11.96401, 78.05134	15 KM/Hr
52	Turn	Medium	11.96504, 78.04975	30 KM/Hr
53	Turn	High	11.96597, 78.04920	15 KM/Hr
54	Turn	Medium	11.96679, 78.04942	30 KM/Hr
55	Turn	Medium	11.96804, 78.05080	30 KM/Hr
56	Turn	Medium	12.16993, 78.16780	30 KM/Hr
57	Turn	Medium	12.19885, 78.18158	30 KM/Hr
58	Turn	High	12.21900, 78.19171	15 KM/Hr
59	Turn	Medium	12.21793, 78.19495	30 KM/Hr
60	Turn	Medium	12.21646, 78.19612	30 KM/Hr
61	Turn	High	12.21363, 78.21154	15 KM/Hr
62	Turn	High	12.21457, 78.21206	15 KM/Hr
63	Turn	Medium	12.21324, 78.21506	30 KM/Hr
64	Turn	High	12.21244, 78.21558	15 KM/Hr
65	Turn	Medium	12.21113, 78.22072	30 KM/Hr
66	Turn	Medium	12.21135, 78.22257	30 KM/Hr
67	Turn	Medium	12.20519, 78.24453	30 KM/Hr
68	Turn	Medium	12.20752, 78.24730	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
9	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
10	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
13	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium
16	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots

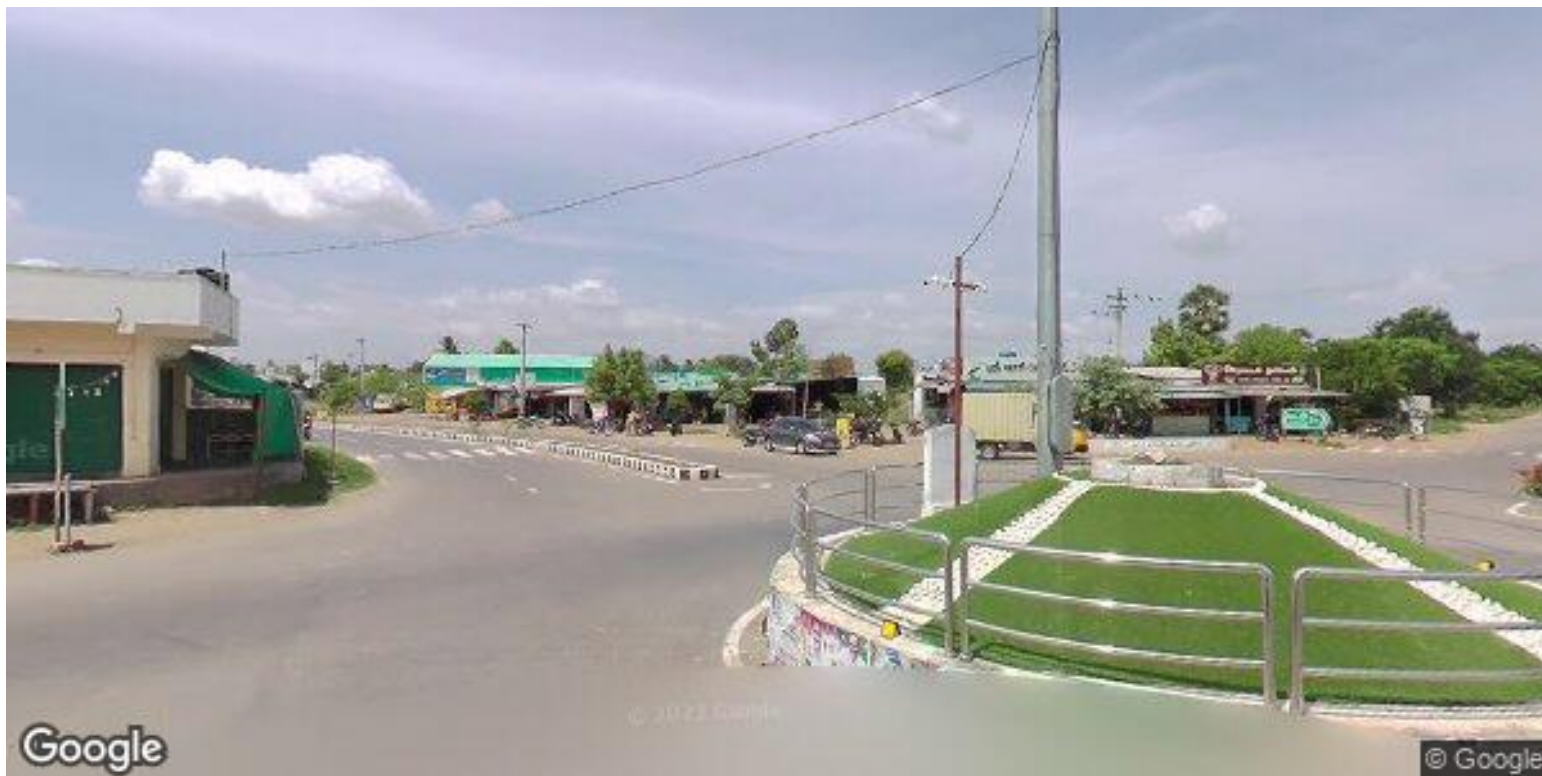


Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60359, 77.83965



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.23740, 78.27029



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49385, 77.86709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49513, 77.86656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50961, 77.86008



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52791, 77.85581



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55685, 77.85813



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56411, 77.85969



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57910, 77.85555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58709, 77.85824



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.60437, 77.83983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61007, 77.84163



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61835, 77.84026



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63147, 77.84083



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64049, 77.84596



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64166, 77.84871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64516, 77.85108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65359, 77.85231



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67319, 77.86041



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67365, 77.86215



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68487, 77.86933



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68478, 77.87285

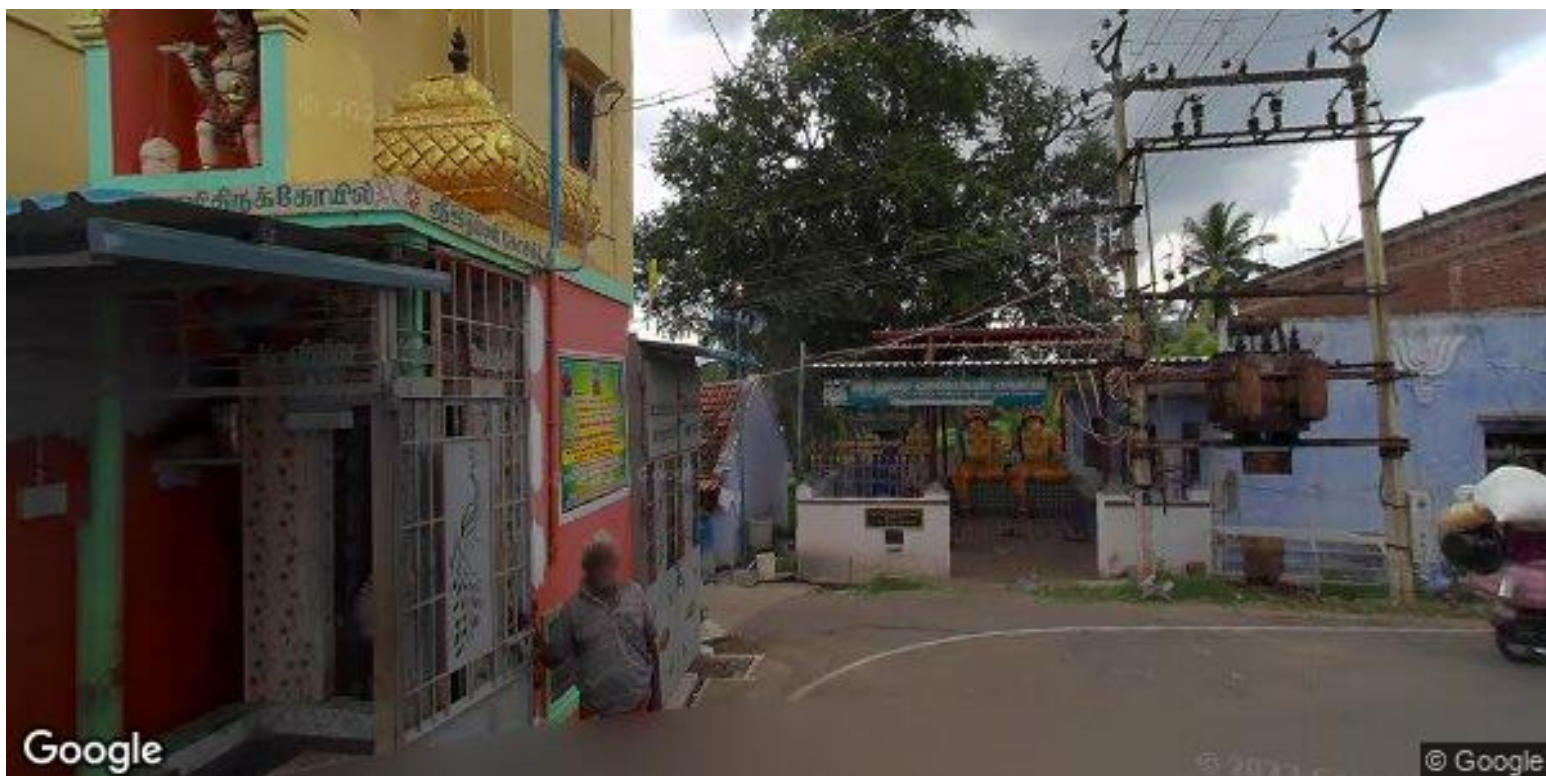


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69463, 77.87589



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69549, 77.87444



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76020, 77.89071



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75960, 77.89221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83330, 77.94432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83370, 77.94485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.84179, 77.94732



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89747, 77.98079



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.90011, 77.98213



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91901, 78.02623



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93618, 78.05392



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93710, 78.05390



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95824, 78.04981



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95992, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96597, 78.04920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96679, 78.04942



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96804, 78.05080



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.16993, 78.16780



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.19885, 78.18158



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21900, 78.19171



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21793, 78.19495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21646, 78.19612



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21363, 78.21154



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21457, 78.21206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21324, 78.21506



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21244, 78.21558



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21135, 78.22257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20519, 78.24453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20752, 78.24730
